Local Income Taxes and Residential Sorting
Evidence from LEHD Origin–Destination Data

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Motivation

- Tiebout (1956): citizens sort across jurisdictions according to their demands for local public goods.
- Strong evidence of tax-driven sorting at the state level (Kleven et al., 2020; Young et al., 2016).
- Little causal evidence on local income tax sorting within metro areas, where moving costs are small and boundaries frequent.
- Two data problems make this hard. Local tax jurisdictions are sub-county, so IRS/ACS migration data are too coarse. Credit systems and weak enforcement complicate the effective tax burden.

This Paper

- Question: Do local income tax rate changes cause sorting?
- Setting: Pennsylvania, 2002–2019. Universal Earned Income Tax with continuous rate variation across ∼2,500 munis × ∼500 school districts.
- Data: LEHD LODES origin–destination, linking residence tract with workplace tract annually.
- Outcomes: `share_live_in` = (workers who live and work in city i) / (total workers in i), plus count outcomes (resident workers, total jobs).
- Exogeneous identification: Act 32 consolidated collection and made employer withholding mandatory.

Institutional Setting (Pennsylvania)

- Pennsylvania has a universal Earned Income Tax under Act 511 of 1965. About 2,500 municipalities and 500 school districts each set their own rate. The typical muni rate is 0.5%; the typical school district rate is also 0.5%.
- Each address has a six-digit PSD code that identifies its (muni, school district) pair for withholding purposes.
- Before Act 32, collection was handled by roughly 560 separate tax officers across the state. Employer withholding was inconsistent, and many munis did not even report a nonresident rate to the state.
- Act 32 of 2008 (effective January 1, 2012) consolidated collection into 69 Tax Collection Districts and required employers to withhold the higher of the home resident rate and the work nonresident rate.
- In our 2002–2019 panel: 210 muni rate increases and 119 decreases; 611 school district increases and 52 decreases.
- Each muni has three rates we track: $\tau^{res,muni}$ on residents, $\tau^{nr,muni}$ on

How the PA EIT works (post-Act 32, excluding Philadelphia)

A worker pays the higher of the total home rate (muni resident plus school district) and the nonresident rate at the work location. The bill is split piecewise:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr}},\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr}} \\ \left(0,\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr}} \end{cases}$$

- The work jurisdiction always collects its full nonresident rate.
- The home muni and school district then split whatever remains of the worker's bill. They get nothing if the work rate is higher.
- Which rate binds determines how the worker can respond. When the home rate binds, the worker can lower the bill by moving home. When the work rate binds, the worker can lower it only by changing jobs.

Worked example

Worker 1 lives in city B and works in city A. Home muni rate is 1.0%, home school district is 0.5%, so the total home rate is 1.5%. The work-location nonresident rate is 0.8%.

- Worker 1 pays the higher of the two rates, so 1.5%. Home collects $1.5 - 0.8 = 0.7\%$; work collects 0.8%.
- This is a home-binds case. To lower the bill, worker 1 would have to move out of B. Changing jobs would not help because the work rate is already lower.

Philadelphia: a separate regime

- Philadelphia is exempt from Act 32. Its Wage Tax operates under the Sterling Act of 1932, administered by the City Department of Revenue.
- Philly residents pay the Philly resident rate regardless of where they work. There is no max rule and no offset.
- Nonresidents working in Philly pay the Philly nonresident rate, which goes entirely to Philly.
- In our analysis we keep Philly in the main sample because it is part of PA's labor market, but exclude it in robustness checks because its reporting and dynamics differ from Act-32 munis.

Data Sources

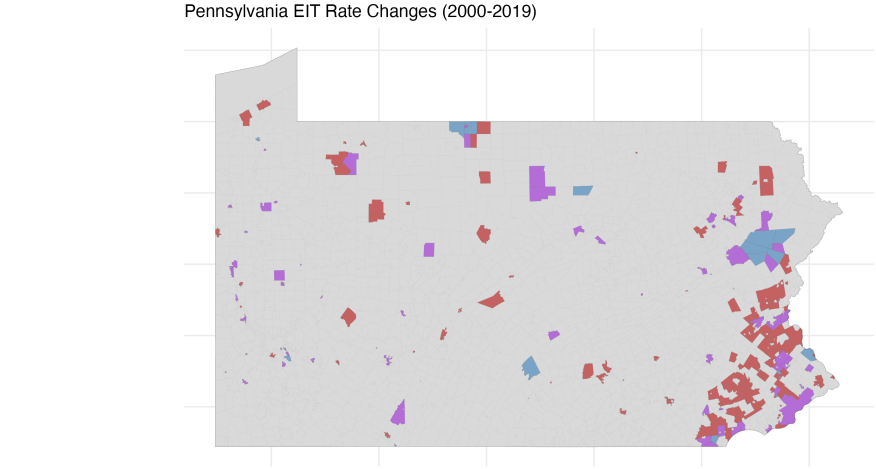
- LODES v8 (2002–2019, trimmed pre-COVID): tract-pair-year OD flows, main + aux files, segments SE01/SE03/SA01-03/JT04.
- PA rates: scraped DCED MunStats (resident + nonresident muni EIT, school district EIT) ∼2,500 munis/yr; name-based COUSUBFP crosswalk (99.6% match).
- PA fiscal controls: DCED Annual Financial Reports (71 cols: EIT rev, police, fire, streets, sewer, parks, comm dev, debt, state aid, pop, MV).

Summary Statistics (Pennsylvania, 2002–2019)

Variable	Mean	Median	SD	Max
share_live_in	0.149	0.127	0.097	0.700
Muni resident EIT (%)	0.542	0.500	0.199	4.331
Muni nonres EIT (%)	0.546	0.500	0.207	4.331
SD EIT (%)	0.523	0.500	0.221	1.000
Total jobs	3,624	1,224	19,248	730,988
Resident workers	674	154	9,507	389,798
Muni × year	27,973 (1,479 munis)			

Notes: Sample restricted to munis with > 50 total jobs. Pre-2012 muni nonres rates backfilled from earliest reported post-2012 value where missing (96% of munis stable).

Treatment Geography: PA Rate Changes



Red = increased; blue = decreased; purple = both; grey = unchanged.

Three Channels of Adjustment

When muni A raises its rate, workers in A can:

1. Move residence only (keep job in A). Under credit system: liability still τ_A. Zero tax savings, but reduces `share_live_in`.
2. Move residence and job to lower-rate B. New liability τ_B. Canonical Tiebout margin.
3. New hires choose to live outside A. Reduces share even without existing-resident moves.

Predictions we'll test: (1) enforcement matters → post-Act 32 effect (Act 32 break); (2) destination rate irrelevant for commuters when work-binds; (3) complete relocators pick lower-rate destinations; (4) SD rate opposite sign (Q5); (5) boundary gradient (Q6).

Q2: Do rate increases cause sorting?

Unit i = PA municipality (cousub); t = year, 2002–2019. Outcome $\texttt{share_live_in}_{it}$ = workers who live and work in muni i at year t, divided by total workers in i. Workers responding pay home stack $\tau_i^{\text{res,muni}} + \tau_i^{\text{SD}}$; nonres rate doesn't enter. Three specs:

$$\text{(A) total stack:} \quad Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{home}} + \varepsilon_{it}$$

$$\text{(B) decomposition:} \quad Y_{it} = \alpha_i + \gamma_t + \beta^{m} \tau_{it}^{\text{res,muni}} + \beta^{sd} \tau_{it}^{\text{SD}} + \varepsilon_{it}$$

$$\text{(C) muni only (back-compat):} \quad Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{res,muni}} + \varepsilon_{it}$$

- Sun–Abraham + DCDH staggered estimators with three cohort definitions: muni-rate event, SD-rate event, total-stack event.
- Fiscal controls: police, fire, streets, sewer, parks, comm dev, gen admin, debt, state aid (full set).

Q2 Answer: muni rate drives the result; SD adds noise

	Muni only	Total stack	Decomp
$\tau^{\text{res,muni}}$	-0.013^{**}		-0.013^{**}
τ^{SD}			$+0.003$
τ^{home}		-0.005	

- Muni rate: -1.3 share-pp per 1 pp hike. SD rate: zero direct effect on `share_live_in`.
- Bundling them dilutes the signal – SD rate is uncorrelated with the sorting margin we measure here.
- SD does have effects on resident worker counts (Q5) with opposite sign from muni. The decomposition is essential.

Mechanism: Is it enforcement? (Act 32)

Unit i = PA municipality; t = year. Outcome $\texttt{share_live_in}_{it}$. Structural break on rate effect:

$$Y_{it} = \alpha_i + \gamma_t + \beta^{\text{pre}}\tau_{it} \cdot \mathbf{1}[t < 2012] + \beta^{\text{post}}\tau_{it} \cdot \mathbf{1}[t \geq 2012] + \varepsilon_{it}$$

- Pre-2012: fragmented collection, weak withholding; DCED's nonresident rate registry was incomplete (only ∼55% of munis reported nonres rates; jumped to ∼72% in 2012).
- Post-2012: centralized 69 TCDs, mandatory employer withholding at $\max(\tau^{\text{home}}, \tau^{\text{nr,work}})$.
- Run break for both rates separately to see which channel is enforcement-bound.

Act 32 Answer: Sorting only once the tax is binding

	Pre-2012	Post-2012
$\tau^{\text{res,muni}}$ (muni only)	-0.003	-0.018^{***}
τ^{home} (total stack)	-0.005	-0.005
$\tau^{\text{res,muni}}$ (decomp)	-0.003	-0.017^{***}
τ^{SD} (decomp)	-0.001	$+0.007$

- Muni rate: precise zero before, -1.8 share-pp after. Difference significant at 1%. Same statutory rate, very different behavioral response.
- Total stack hides the break: SD rate did not get an Act-32 enforcement shock (state-administered withholding existed already).
- Decomposition: enforcement break is on the **muni channel only**. SD plays a different role (Q5).

Q3: Are rate changes symmetric?

Unit i = PA municipality; t = year. Outcome $\texttt{share_live_in}_{it}$. Split cumulative dose from each muni's baseline (first observed) rate:

$$Y_{it} = \alpha_i + \gamma_t + \beta^+ R_{it}^+ + \beta^- R_{it}^- + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

- R_{it}^+, R_{it}^-: cumulative increases / decreases from baseline.
- If $|\beta^+| > |\beta^-| \rightarrow$ residential choices are sticky (fixed costs of moving).
- Also estimate joint inc/dec event indicators.

Q3 Answer: Sharply asymmetric on muni rate

	R^+ (cum. inc)	R^- (cum. dec)
Muni only	-0.018^{**}	$+0.002$
Total stack	-0.006	-0.001
Decomp: muni	-0.018^{***}	$+0.001$
Decomp: SD	$+0.003$	$+0.019$

- Asymmetry on **muni rate only**: increases push out (-1.8 share-pp per pp), decreases do not bring back.
- SD rate: no asymmetry on share_live_in (SD effect lives on count outcomes – Q5).
- Policy: a city that raises then lowers its muni rate will not recover the residents it lost.

Q4: Where do displaced residents go?

Two units, two specs:

- **(a) Resident workers**: unit B = PA municipality; t = year. Outcome = log(workers living in B, any workplace). Full panel; home channel only.

$$\ln(\text{ResW}_{Bt}) = \alpha_B + \gamma_t + \beta_1 \tau_{Bt}^{\text{home}} + \beta_2 \bar{\tau}_{Bt}^{\text{nbr}} + \varepsilon$$

- **(b) Commuter flow**: unit = ordered muni-pair (A, B) with $A \neq B$, within 80 km; t = year. Outcome = log(workers commuting from home B to work A). Post-Act-32 only.

$$\ln(\text{Flow}_{ABt}) = \alpha_{AB} + \gamma_t + \beta_W \tau_{At}^{\text{nr}} + \beta_H \tau_{Bt}^{\text{home}} + \varepsilon$$

Predictions: $\beta_2 > 0$ in (a) – neighbors absorb displaced residents; $\beta_H \approx 0$ in (b) when work binds (credit system makes home rate moot for commuters).

Q4 Answer: Residents leave; neighbors absorb

	log(Resident Workers) home stack on B, full panel	Commuter flow $B \to A$ post-2012, piecewise
Own muni $\tau_B^{\text{res,muni}}$	-0.122^{**}	$+0.054^{***}$
Own SD τ_B^{SD}	$+0.147^{***}$	≈ 0
Neighbor avg	$+1.06^{*}$	—
Work nonres τ_A^{nr}	—	$+0.019$ (n.s.)

- Resident workers in B: muni rate repels (-12% per pp); SD rate attracts ($+15\%$). *Neighbors absorb* the displaced ($\beta = +1.06^{*}$).
- Neighbor = within 80 km centroid distance (regional commute shed).
- Commuter flow $B \to A$: work nonres rate near zero (small, n.s.). Home muni rate positive on flow – a *selection signature*, not causal: high-rate home cities house residents with stronger commuting margins.
- Both panels much stronger post-Act-32 (resident-workers muni effect $5\times$ larger).

Q5: Muni vs SD — decomposition across outcomes

Unit i = PA municipality; t = year. Decomposition $\beta^{m}\tau_{it}^{\text{res,muni}} + \beta^{sd}\tau_{it}^{\text{SD}}$ estimated for each of four outcomes. Q4 looked at one outcome (resident worker count) with neighbor exposure; Q5 fixes the decomposition spec and varies the outcome to identify which workers respond to which rate.

- Outcomes: `share_live_in`, log(resident workers), log(nonresident workers), log(total jobs).
- Theory: SD only taxes residents $\rightarrow \beta^{sd} \approx 0$ on nonresident outcome but nonzero on resident outcomes. Muni res rate same direction across all but smaller for nonres workers (who pay only $\tau^{\text{nr,muni}}$, which is correlated but not identical).
- Muni–SD rate correlation $r = -0.12 \rightarrow$ independent variation.

Q5 Answer: Opposite signs — SD attracts residents

	share_live_in	log(Resident)	log(Nonres.)	log(Total)
$\tau^{\text{res,muni}}$	-0.010^{**}	-0.117^{**}	-0.078^{*}	-0.094^{**}
τ^{SD}	$+0.002$	$+0.141^{***}$	$+0.077$	$+0.076$
Total stack	~ 0 on all – hides both channels			

- Muni rate consistently negative; SD strongly positive on residents, ~ 0 on nonresidents (theory: SDs don't tax commuters).
- Bundling them as "total EIT" nets to zero → prior literature using a single rate would find a null.
- Decomposition is the methodological contribution.

Robustness

- Pre-trends clean across 11 observables: revenue, EIT rev, RE tax, streets, pop, debt, jobs, crime, permits. Exceptions (police, market value) addressed with controls / coverage notes.
- Fiscal controls (log police, streets, debt, state aid) leave point estimates and SEs essentially unchanged.
- Philadelphia exclusion: results robust to dropping Philly.
- Flip-floppers: 37% of PA treated munis flip-flop; excluding them strengthens results.
- Placebo: Federal workers (JT04) show no response (federal wages not subject to local withholding same way).
- ACS / IRS SOI: county-level migration null — LODES detects what coarser data cannot.

Putting it Together (1/2): Who moves?

Setup. A post-Act 32 PA city: 1,000 total workers; 150 live-and-work there (share_live_in = 15%); EIT rises 0.5% → 1.5%.

Q2/Q3. Over 7–8 years, share falls to ∼13.2% → ∼18 of 150 locally-resident workers gone. If the city later cuts the rate back, none of them return.

Q4. Those 18 leave the city entirely (not a neighborhood move); nearby lower-rate munis within 80 km absorb them.

Q5. If the overlapping school district had raised its rate instead, residents would have moved in, not out.

Q6. The loss is spread across the city, not concentrated at municipal boundaries — no "edge effect."

Putting it Together (2/2): Was it worth it?

Same city: 1,000 workers, rate 0.5% → 1.5%, ∼18 residents lost.

Fiscal. EIT revenue roughly doubles (+67 log points); total tax +30%. Property values and population: unchanged.

Q7. Most new dollars flow to community development (planning, econ dev, housing) — not to police patrols residents see.

nforcement. None of this happens pre-Act 32: same statutory rate, precise zero behavioral response. Collection capacity is the binding constraint.

Bottom line. The city trades ∼18 residents for a near doubling of EIT revenue. Rate increases work — when enforced.

Summary of Findings

1. Higher muni rates reduce `share_live_in` by about 1.3 share-points per 1 pp, building over 7 to 8 years.
2. The effect is enforcement-contingent. The muni rate coefficient is a precise zero before Act 32 and roughly -1.8 share-points after.
3. The response is asymmetric. Increases push residents out, but decreases do not bring them back.
4. Muni and school district rates have opposite signs. Bundling them as a single "total EIT" rate hides both channels.
5. Sorting is whole-city relocation, not boundary arbitrage.

Next Steps

- Link Act 1 SD rate changes to actual property tax millage movements (PDE data 2015+ restrictive).
- Extend Q7 with richer, pre-treatment expenditure composition (more power on visibility channel).
- Apply the LODES-based design to other local instruments (property tax millage, local sales taxes, fees).
- Welfare: combine behavioral elasticity with revenue elasticity to bound efficiency cost of local income tax.

Thank You

Questions?

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github.com/MatthewHockert/local-income-tax-migration

References I

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